

KISAN INJECTION 10MG/ML

WARNING - INTRAVENOUS USE

Severe reactions, including fatalities, have occurred during and immediately after the parenteral administration of KISAN INJECTION (Phytonadione). Typically these severe reactions have resembled hypersensitivity or anaphylaxis, including shock and cardiac and/or respiratory arrest. Some patients have exhibited these severe reactions on receiving KISAN INJECTION for the first time. The majority of these reported events occurred following intravenous administration, even when precautions have been taken to dilute the KISAN INJECTION and to avoid rapid infusion. Therefore, the INTRAVENOUS route should be restricted to those situations where another route is not feasible and the increased risk involved is considered justified. When slow i.v injection is given, the rate should not exceed 1mg/min.

DESCRIPTION: Kisan Injection is a clear golden yellow liquid.

COMPOSITION: Each 1 ml ampoule contains Vitamin K1 (Phytomenadione) 10 mg.

Preservative: Benzyl Alcohol 9.0 mg/ml. **Solubilizing Agent:** Cremophor EL (polyoxyethylated castor oil)

PHARMACODYNAMICS: Class: (Vitamin K1).

Site & Mode of Action: As a component of an enzyme system, Vitamin K1 promotes the formation in the liver of coagulation factors II (prothrombin), VII, IX and X. Anticoagulants of the coumarin and indandione series cause a reversible displacement of Vitamin K1 from this enzyme system, thereby inhibiting the synthesis of these factors. Since this is a competitive displacement, Vitamin K1 is a specific antagonist for warfarin and similar anticoagulants. It is not capable, however, of terminating the action of heparin; for this purpose a salt of protamine should be used.

PHARMACOKINETICS: Orally ingested phytomenadione is absorbed primarily in the middle portions of the small intestine. Optimum absorption is possible in the presence of bile and pancreatic juice. Systemic availability after oral dosing is about 50%, with a wide range of interindividual variability. The primary distribution compartment corresponds to the plasma volume. In blood plasma, 90% of Vitamin K1 is bound to lipoprotein (VLDL portion). Vitamin K1 plasma concentration is normally between 0.4 and 1.2 mcg/L. Vitamin K1 does not readily cross the placenta and is poorly distributed into breast milk.

The elimination half-life in plasma is 1.5 – 3 hours. After metabolic degradation Vitamin K1 is bound to glucuronic acid and excreted in the bile and urine.

Less than 10% of the drug is excreted unchanged in the urine. Vitamin K1 has an active metabolite, vitamin K1-2, 3 epoxide, which is reconverted into Vitamin K1. Impaired absorption may occur in conditions such as malabsorption syndromes, short bowel syndrome, biliary atresia and pancreatic insufficiency.

INDICATIONS: Haemorrhage or threatened haemorrhage as a result of severe "hypoprothrombinemia" (i.e. deficiency of coagulation factors II, VII, IX and X) due, for instance, to overdosage of anticoagulants of the dicoumarol type or their combination with phenylbutazone, or to other forms of hypovitaminosis K (e.g. obstructive jaundice, liver and intestinal disorders, or prolonged administration of antibiotics, sulfonamides or salicylates).

RECOMMENDED DOSAGE:

Vitamin K1 may be given orally or slow i.v. injection (10 mg ampoules). Slow IV injection must be reserved for potentially fatal haemorrhage due to overdosage of anticoagulants of the coumarin and indandione series. Single doses of 20 mg and total doses of 40 mg should be regarded as the maximum. Excessive doses impede the resumption of anticoagulant therapy without offering any advantages.

In severe bleeding, or situations where other routes are not feasible, Vitamin K1 may be given by very slow intravenous injection, at a rate not exceeding 1 mg per minute.

Mild Haemorrhage: Can usually be controlled by oral doses of 10 mg Vitamin K1. If the prothrombin level dose does not rise sufficiently within 8-12 hours or haemorrhage continues, a further dose, increased if necessary, should be given.

In Severe Haemorrhage due to Hypoprothrombinaemia: (arising, for example, during treatment with coumarin derivatives), 10-20 mg of Vitamin K1 injection (1-2 ampoules 10 mg) by slow i.v. injection. The prothrombin level should be estimated 3 hours later and if the response has been inadequate the dose should be repeated. If circumstances require, blood transfusions should be in addition to Vitamin K1.

If surgical intervention should be necessary in a patient receiving anticoagulants of the coumarin or indandione series, their anticoagulant action, may if necessary be counteracted by means of Vitamin K1. If there is recurrence of thrombosis while Vitamin K1 is being used IV administration of heparin is recommended as a first measure.

Geriatric: Elderly patients tend to be more sensitive to reversal of anticoagulation with vitamin K1. The dosage for this patient group should therefore be at the lower end of the ranges recommended.

In Impaired Liver and Renal Function: In severe liver disease, Vitamin K1 should be discontinued if no significant effect is noted within 1-2 days after the initial dose.

ROUTE OF ADMINISTRATION: For Slow I.V. or Oral Use only

CONTRAINDICATIONS: Vitamin K1 ampoules should not be used for patients with pronounced allergic diathesis or with a known history of a severe hypersensitivity reaction to agents containing Cremophor® EL or its derivatives such as polyoxyethylated castor oil. Kisan Injection 10mg/ml should not be administered intramuscularly because the i.m. route exhibits depot characteristic and continued release of vitamin K1 would lead to difficulties with the re-institution of anticoagulation therapy. Furthermore, i.m. injections given to anticoagulated subjects cause a risk of haematoma formation.

WARNING & PRECAUTIONS:

Warning:

As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age. Not to be used in neonates.

Vitamin K1 should be considered as adjunctive therapy to blood transfusions for severe haemorrhage due to anticoagulant therapy; it is not effective when heparin-like compounds have been used for anticoagulant therapy; minimal doses should be used to offset refractoriness to coumarin-like anticoagulants if long term anticoagulant therapy is intended. Severe reactions, including fatalities, have occurred during and immediately after intravenous injection of Vitamin K1. Restrict intravenous use of emergency cases. When intravenous administration is necessary the rate of injection should not exceed 1 mg per minute.

INTERACTION WITH OTHER MEDICATIONS:

- Concurrent use of anticoagulants, coumarin or indandione-derivative with Vitamin K may decrease the effects of these anticoagulants as a result of increased hepatic synthesis of pro-coagulant factors.
- Requirements for Vitamin K may be increased in patients receiving broad-spectrum antibiotics, Quinidine, Quinine, high doses of salicylates or antibacterial sulphonamides.

USE IN PREGNANCY & LACTATION: Use in pregnancy: There is no specific evidence regarding the safety of Vitamin K1 in pregnancy and no reproductive studies have been performed in animals. Therefore, as with most drugs, administration during pregnancy should only occur if the benefits outweigh the risks. Not recommended for pregnant women as prophylaxis of vitamin K deficiency bleeding in the newborn.

SIDE EFFECTS: Intravenous injection of Vitamin K1 may, in rare cases, cause severe, anaphylactoid reactions. Should an anaphylactoid reaction occur, the usual measures must be taken (eg. administration of adrenaline and supportive measures as required).

SYMPTOMS AND TREATMENT OF OVERDOSE: Overdosage: Hypervitaminosis of Vitamin K1 has not been reported.

STORAGE CONDITIONS: Store below 30°C. Protect from light. KEEP OUT OF REACH OF CHILDREN. *JAUHKAN DARIPADA KANAK-KANAK.* DO NOT ALLOW TO FREEZE. IF SEPARATION HAS OCCURRED OR IF OIL DROPLETS HAVE APPEARED, THE INJECTION SHOULD NOT BE USED.

PACK SIZE: Packs of 10 ampoules x 1 ml

SHELF LIFE: Please refer outer package

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
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